RESEARCH MAP · 2026

可控语音合成：从 Model 到 Agent

Controllable Speech Synthesis: From Generative Models to Interactive Voice Agents

范围：可控 TTS、上下文自适应语音生成、Agentic Speech Production、Interactive Voice Agent；重点优先整理具有规划、检索、工具、批评反馈、流式状态或实时行动机制的工作。

更新时间：2026-08-15。论文按首次公开时间排序；同一工作若同时贡献数据集、训练方法或评测，可在对应证据层交叉列出。

核心判断

- **控制对象发生变化：**从音高、韵律、音色和情绪，扩展到角色、对话行为、话轮时机、工具调用与任务行动。
- **控制来源发生变化：**从用户显式 Prompt，转向系统基于语音、上下文、Persona、目标和反馈自主推断。
- **Agent 的分界不在名称：**只有当规划、检索、批评或在线状态决策真正影响语音生成/交互时，才进入 Agent 主线。
- **2026 年的关键跃迁：**AgentSteerTTS、Harness TTS、ActorMind、PRISM、DuplexSLA 等开始把可控语音纳入闭环决策。

工作分类与判断边界

Level	Control decision	Agent mechanism	Role of speech	Inclusion boundary
M · Model	User/reference explicitly supplies emotion, prosody, timbre, voice or style.	None required; model executes a condition.	Generated artifact	Controllable TTS anchor; not an agent.
C · Context-aware	System infers how to speak from user speech, history, affect, persona or goal.	Context inference; optional retrieval or emotion reasoning.	Context-aligned response	Agent-adjacent unless an explicit decision loop exists.
A · Agentic Speech	Planner chooses style/content/voice and may revise it.	Planning, retrieval/tool routing, critic, feedback or multi-agent orchestration.	Planned/revised speech artifact	Agent loop acts on speech generation, often offline.
V · Voice Agent	System decides how and when to speak while interacting online.	Streaming state, reasoning, tool/action, turn policy and feedback.	Real-time action	Speech participates in a live reason-plan-act loop.

符号说明：M = Model-level；C = Context-aware；A = Agentic Speech；V = Interactive Voice Agent。该四级划分是本调研使用的工作分类，不宣称为社区统一标准。

1. Surveys and Positioning

直接综述用于界定可控属性、模型架构和评测；邻近综述用于补足从语音模型到全双工交互的系统脉络。

1. **"Towards Controllable Speech Synthesis in the Era of Large Language Models: A Survey".** *Tianxin Xie et al.* EMNLP 2025 / arXiv 2024. [Paper]
2. **"WavChat: A Survey of Spoken Dialogue Models".** *Shengpeng Ji et al.* arXiv 2024. [Paper]
3. **"From Turn-Taking to Synchronous Dialogue: A Survey of Full-Duplex Spoken Language Models".** *Yuxuan Chen and Haoyuan Yu* arXiv 2025. [Paper]
4. **"A Survey of Full-Duplex Spoken Dialogue Systems: Architectural Hierarchy, Interaction Ontology, and Decision State Machine".** *Jingyu Lu et al.* arXiv 2026. [Paper]

2. Model-Level Controllable TTS（模型级可控合成）

这一层回答"给定控制条件，模型能否稳定地产生目标语音"。它构成 Agent 的执行器基础，但控制决策仍主要来自用户、参考语音或预先定义的 Prompt。

1. **"Style Tokens: Unsupervised Style Modeling, Control and Transfer in End-to-End Speech Synthesis".** *Yuxuan Wang et al.* arXiv 2018. [Paper]
2. **"Mellotron: Multispeaker Expressive Voice Synthesis by Conditioning on Rhythm, Pitch and Global Style Tokens".** *Rafael Valle et al.* arXiv 2019. [Paper]
3. **"PromptTTS: Controllable Text-to-Speech with Text Descriptions".** *Zhifang Guo et al.* arXiv 2022. [Paper]
4. **"InstructTTS: Modelling Expressive TTS in Discrete Latent Space with Natural Language Style Prompt".** *Dongchao Yang et al.* arXiv 2023. [Paper]
5. **"NaturalSpeech 2: Latent Diffusion Models are Natural and Zero-Shot Speech and Singing Synthesizers".** *Kai Shen et al.* arXiv 2023. [Paper]
6. **"PromptTTS 2: Describing and Generating Voices with Text Prompt".** *Yichong Leng et al.* ICLR 2024 / arXiv 2023. [Paper]
7. **"Natural Language Guidance of High-Fidelity Text-to-Speech with Synthetic Annotations".** *Dan Lyth and Simon King* arXiv 2024. [Paper]
8. **"NaturalSpeech 3: Zero-Shot Speech Synthesis with Factorized Codec and Diffusion Models".** *Zeqian Ju et al.* arXiv 2024. [Paper]
9. **"ControlSpeech: Towards Simultaneous Zero-shot Speaker Cloning and Zero-shot Language Style Control With Decoupled Codec".** *Shengpeng Ji et al.* arXiv 2024. [Paper]
10. **"CosyVoice: A Scalable Multilingual Zero-shot Text-to-speech Synthesizer based on Supervised Semantic Tokens".** *Zhihao Du et al.* arXiv 2024. [Paper]
11. **"CosyVoice 3: Towards In-the-wild Speech Generation via Scaling-up and Post-training".** *Zhihao Du et al.* arXiv 2025. [Paper]
12. **"Qwen3-TTS Technical Report".** *Qwen Team et al.* arXiv 2026. [Paper]

3. Context-Aware Conversational Speech Synthesis（上下文自适应）

这一层开始回答"在当前对话中应该怎么说"。系统利用用户语音、历史、情绪、Persona 或检索到的相似对话推断表达方式，但并非每项工作都具有完整 Agent 闭环。

1. **"Talk With Human-like Agents: Empathetic Dialogue Through Perceptible Acoustic Reception and Reaction".** *Haoqiu Yan et al.* arXiv 2024. [Paper]
2. **"Style-Talker: Finetuning Audio Language Model and Style-Based Text-to-Speech Model for Fast Spoken Dialogue Generation".** *Yinghao Aaron Li et al.* arXiv 2024. [Paper]
3. **"JELLY: Joint Emotion Recognition and Context Reasoning with LLMs for Conversational Speech Synthesis".** *Jun-Hyeok Cha et al.* arXiv 2025. [Paper]
4. **"Retrieval-Augmented Dialogue Knowledge Aggregation for Expressive Conversational Speech Synthesis".** *Rui Liu et al.* arXiv 2025. [Paper]
5. **"Chain-Talker: Chain Understanding and Rendering for Empathetic Conversational Speech Synthesis".** *Yifan Hu et al.* arXiv 2025. [Paper]
6. **"Voicing Personas: Rewriting Persona Descriptions into Style Prompts for Controllable Text-to-Speech".** *Yejin Lee et al.* arXiv 2025. [Paper]
7. **"EmoNews: A Spoken Dialogue System for Expressive News Conversations".** *Ryuki Matsuura et al.* arXiv 2025. [Paper]

8. **"ES4R: Speech Encoding Based on Prepositive Affective Modeling for Empathetic Response Generation".** *Zhuoyue Gao et al.* arXiv 2026. [Paper]
9. **"Sympatheia: Emotionally Adaptive Voice Assistant with Continuous Affect Conditioning".** *Sukru Samet Dindar et al.* arXiv 2026. [Paper]
10. **"Self-EmoQ: Plutchik-Guided Value-based Planning to Drive Streaming Emotional TTS".** *Yue Zhao et al.* arXiv 2026. [Paper]

4. Agentic Speech Generation and Orchestration（重点）

本节优先收录存在显式 **Planner**、**Retriever**、**Critic**、**Feedback** 或 **Multi-Agent** 协作，并且这些机制真实控制语音生成、角色配音或音频编排的工作。

1. **"PodAgent: A Comprehensive Framework for Podcast Generation".** *Yujia Xiao et al.* arXiv 2025. [Paper]

   定位：Host–Guest–Writer 多智能体协作，联合内容规划、角色配音和表达性语音生成；属于离线 Agentic Speech Production。

2. **"DialogueAgents: A Hybrid Agent-Based Speech Synthesis Framework for Multi-Party Dialogue".** *Xiang Li et al.* ICME 2025 / arXiv 2025. [Paper]

   定位：Script Writer、Speech Synthesizer 与 Dialogue Critic 构成迭代闭环，并产出 MultiTalk；Agent 作用于脚本与语音质量控制。

3. **"ActorMind: Emulating Human Actor Reasoning for Speech Role-Playing".** *Xi Chen et al.* arXiv 2026. [Paper]

   定位：Eye–Ear–Brain–Mouth 多 Agent 推理链把角色、场景和情绪状态转为个性化语音表演。

4. **"AuDirector: A Self-Reflective Closed-Loop Framework for Immersive Audio Storytelling".** *Yiming Ren et al.* arXiv 2026. [Paper]

   定位：角色规划、语音检索、合成审查、自我修正与人类反馈共同形成长音频制作闭环。

5. **"AgentSteerTTS: A Multi-Agent Closed-Loop Framework for Composite-Instruction Text-to-Speech".** *Bin Kang et al.* arXiv 2026. [Paper]

   定位：Retrieval Agent、Synthesis Agent 与 Fast–Slow Feedback Agent 共同解决复合指令的语义–声学对齐。

6. **"Audio-Oscar: A Multi-Agent System for Complex Audio Scene Generation, Orchestration, and Refinement".** *Yifan Duan et al.* arXiv 2026. [Paper]

   定位：面向复杂音频场景的规划、角色/声线设计、语音与非语音生成、时间线编排和反馈式精修。

7. **"PRISM: Prosody-Integrated Multi-Agent Reasoning Framework for Empathetic Spoken Dialogue".** *Wen Zhang et al.* arXiv 2026. [Paper]

   定位：将韵律感知、响应生成和语音合成解耦为协作 Agent，并按需调用外部知识工具。

8. **"Harness TTS: Towards Context-Aware Expressive Speech Synthesis with Harness Layer".** *Shengfan Shen et al.* arXiv 2026. [Paper]

   定位：LLM Planner 根据显式、隐式和冲突上下文路由风格 Prompt Tool，提供可审计、低延迟的 TTS 控制层。

5. Interactive Voice Agents（重点）

这一层把语音从"生成结果"提升为"实时行动"：系统需要持续监听、决定何时说、是否打断或 Backchannel，并将推理、角色、工具或结构化 Action 与语音同步。

1. **"F-Actor: Controllable Conversational Behaviour in Full-Duplex Models".** *Maike Züfle et al.* arXiv 2026. [Paper]
   定位：把声线、话题、Backchannel、打断和发起行为统一为可指令控制的全双工对话行为。
2. **"LTS-VoiceAgent: A Listen-Think-Speak Framework for Efficient Streaming Voice Interaction via Semantic Triggering and Incremental Reasoning".** *Wenhao Zou et al.* arXiv 2026. [Paper]
   定位：语义触发器与 Thinker/Speaker 双角色编排，使推理在用户尚未说完时增量展开。
3. **"PersonaPlex: Voice and Role Control for Full Duplex Conversational Speech Models".** *Rajarshi Roy et al.* arXiv 2026. [Paper]
   定位：将角色 Prompt 与声线克隆注入全双工模型；实时角色控制突出，但工具与长程规划仍有限。
4. **"DuplexSLA: A Full-Duplex Spoken Language Model with Synchronized Speech, Language, and Action".** *Haoyang Zhang et al.* arXiv 2026. [Paper]
   定位：在同一 160 ms 时钟上联合解码用户音频、助手语音和结构化 Action，实现边说边规划与工具调用。
5. **"Adaptive Turn-Taking for Real-time Multi-Party Voice Agents".** *Soumyajit Mitra et al.* Interspeech 2026 / arXiv 2026. [Paper]
   定位：ModeratorLM 根据分配角色和多人上下文自主决定是否接管话轮，并提供 CoT 推理变体。
6. **"VoxMind: An End-to-End Agentic Spoken Dialogue System".** *Tianle Liang et al.* ACL 2026. [Paper]
   定位：端到端语音智能体主线，将语音交互、推理和任务执行作为统一系统评估。
7. **"JoyAI-Talker: Full-Duplex Speech Interactive Large Model Built for Empathetic Voice Agents".** *Yinhao Bai et al.* arXiv 2026. [Paper]
   定位：Thinker–Talker、Persona-Adaptive Empathetic Response 与 Joy-Duplex 共同控制内容、声学表达和实时对话状态。

6. Supporting Full-Duplex / Omni Foundations

以下工作提供实时、流式或全双工基础能力，但仅凭该能力不能自动归为 Agent；它们主要构成 Voice Agent 的系统底座。

1. **"Beyond Turn-Based Interfaces: Synchronous LLMs as Full-Duplex Dialogue Agents".** *Bandhav Veluri et al.* EMNLP 2024. [Paper]
2. **"Moshi: A Speech-Text Foundation Model for Real-Time Dialogue".** *Alexandre Défossez et al.* arXiv 2024. [Paper]
3. **"Qwen2.5-Omni Technical Report".** *Qwen Team et al.* arXiv 2025. [Paper]
4. **"MiniCPM-o 4.5: Towards Real-Time Full-Duplex Omni-Modal Interaction".** *Junbo Cui et al.* arXiv 2026. [Paper]

7. Training, Alignment and Reward

从 Model 到 Agent 的另一条主线是由一次性监督转向偏好、奖励、用户交互反馈和多维交互行为优化。

1. **"SpeechAlign: Aligning Speech Generation to Human Preferences".** *Dong Zhang et al.* arXiv 2024. [Paper]
2. **"WavReward: Spoken Dialogue Models With Generalist Reward Evaluators".** *Shengpeng Ji et al.* arXiv 2025. [Paper]
3. **"Aligning Spoken Dialogue Models from User Interactions".** *Anne Wu et al.* arXiv 2025. [Paper]
4. **"Multi-Faceted Interactivity Alignment in Full-Duplex Speech Models".** *Atsumoto Ohashi et al.* arXiv 2026. [Paper]

8. Datasets and Evaluation Benchmarks

评测也从 MOS、相似度和 Prompt Following，扩展到上下文一致性、角色遵循、工具正确性、任务完成、打断鲁棒性和实时交互。

1. **"DailyTalk: Spoken Dialogue Dataset for Conversational Text-to-Speech".** *Keon Lee et al.* arXiv 2022. [Paper]
2. **"SpeechCraft: A Fine-grained Expressive Speech Dataset with Natural Language Description".** *Zeyu Jin et al.* arXiv 2024. [Paper]
3. **"DialogueAgents: A Hybrid Agent-Based Speech Synthesis Framework for Multi-Party Dialogue (MultiTalk)".** *Xiang Li et al.* ICME 2025. [Paper]
4. **"VoiceAgentBench: Are Voice Assistants Ready for Agentic Tasks?".** *Dhruv Jain et al.* arXiv 2025. [Paper]
5. **"ActorMind: Emulating Human Actor Reasoning for Speech Role-Playing (ActorMindBench)".** *Xi Chen et al.* arXiv 2026. [Paper]

6. **"Adaptive Turn-Taking for Real-time Multi-Party Voice Agents (RolePlayConv)".** *Soumyajit Mitra et al.* Interspeech 2026. [Paper]
7. **"Audio-Oscar: A Multi-Agent System for Complex Audio Scene Generation, Orchestration, and Refinement (ASG-Bench)".** *Yifan Duan et al.* arXiv 2026. [Paper]
8. **"τ-Voice: Benchmarking Full-Duplex Voice Agents on Real-World Domains".** *Soham Ray et al.* arXiv 2026. [Paper]
9. **"Full-Duplex-Bench-v3: Benchmarking Tool Use for Full-Duplex Voice Agents Under Real-World Disfluency".** *Guan-Ting Lin et al.* arXiv 2026. [Paper]
10. **"DuplexWorld: Can Voice Agents Help You Get Through the Day?".** *Aryan Vijay Bhosale et al.* arXiv 2026. [Paper]

9. Representative Model-to-Agent Comparison

Table 1 compares representative works along decision source and agent mechanisms. ✓ = native and explicitly evaluated; △ = partial, external, or not the primary contribution; – = not central. Level colors follow the M/C/A/V taxonomy.

Method	Level	Control source	Context	Plan / Reason	Retrieval / Tool	Feedback / Critic	Emotion / Style	Persona / Voice	Real-time / Duplex	Speech as Action	Release
GST-Tacotron	M	Reference / latent token	-	-	-	-	Style, speed	Voice/style	-	-	2018.03
PromptTTS	M	Natural-language prompt	-	-	-	-	Style, prosody	Voice description	-	-	2022.11
InstructTTS	M	Natural-language style instruction	-	-	-	-	Emotion, style	△	-	-	2023.01
ControlSpeech	M	Reference speech + style prompt	-	-	-	-	Style, timbre	✓	-	-	2024.06
Qwen3-TTS	M	Voice sample / text description	-	-	-	-	Fine-grained style	✓	✓	-	2026.01
PerceptiveAgent	C	User speech + dialogue context	✓	△	-	-	Empathic style	△	△	△	2024.06
Style-Talker	C	Audio + history + style	✓	△	-	-	Speaking style	△	✓	△	2024.08
JELLY	C	Speech emotion + dialogue history	✓	✓	-	-	Emotion	-	-	△	2025.01
RADKA-CSS	C	Current + stored dialogues	✓	△	✓	-	Style, empathy	-	-	△	2025.01
Chain-Talker	C	Dialogue history	✓	✓	-	-	Emotion, empathy	-	-	△	2025.05
Sympatheia	C	User affect / VA signal	✓	△	-	-	Continuous affect	-	△	△	2026.06
Self-EmoQ	C	Dialogue context	✓	✓	-	✓	Planned emotion	-	✓	△	2026.06
PodAgent	A	Topic + host/guest roles	✓	✓	✓	△	Role, expressivity	✓	-	△	2025.03
DialogueAgents	A	Character pool + script	✓	✓	△	✓	Emotion, paralinguistics	✓	-	△	2025.04
ActorMind	A	Role + scene + spoken context	✓	✓	-	△	Emotion, verbal traits	✓	-	△	2026.04
AuDirector	A	Narrative + human feedback	✓	✓	✓	✓	Emotion, voice casting	✓	-	△	2026.05
AgentSteerTTS	A	Composite instruction	△	✓	✓	✓	Emotion, intensity	✓	-	△	2026.05
Audio-Oscar	A	Complex scene description	✓	✓	✓	✓	Voice, scene timing	✓	-	△	2026.06
PRISM	A	Prosody + dialogue context	✓	✓	✓	△	Empathic prosody	△	△	✓	2026.06
Harness TTS	A	Explicit / implicit / conflict context	✓	✓	✓	△	Prompt-tool style	✓	✓	△	2026.07
F-Actor	V	Instruction + live context	✓	△	-	-	Voice, behavior	✓	✓	✓	2026.01
LTS-VoiceAgent	V	Streaming user speech + task	✓	✓	△	△	Timing	-	✓	✓	2026.01
PersonaPlex	V	Role prompt + voice sample	✓	△	-	-	Role, style	✓	✓	✓	2026.02
DuplexSLA	V	Audio + action channel	✓	✓	✓	△	Timing, backchannel	△	✓	✓	2026.05
ModeratorLM	V	Role + multi-party context	✓	✓	-	-	Turn behavior	✓	✓	✓	2026.06
VoxMind	V	Spoken task + dialogue state	✓	✓	✓	△	Contextual speech	△	✓	✓	2026.07
JoyAI-Talker	V	Non-verbal cues + persona	✓	✓	△	△	Emotion, rate, volume	✓	✓	✓	2026.08

Interpretation. The decisive shift is not simply from cascaded to end-to-end synthesis. It is the movement of control ownership from the user-supplied condition to a context-sensitive planner, and finally to an online agent that couples speech with timing, tools and actions.

10. Development Trajectory: From Controllable TTS Models to Voice Agents

The trajectory separates six capability phases. Papers are selected from the preceding collection; the table describes a qualitative research mainline rather than a leaderboard.

Phase	Period	Representative milestones	Control shift	Agent mechanism	Open boundary
Latent attribute control	2018-2019	GST-Tacotron; Mellotron	Acoustic/style factors are exposed as controllable latent variables.	No autonomous decision; the user or reference audio supplies control.	Control is interpretable only indirectly and remains utterance-local.
Natural-language control	2022-2024	PromptTTS; InstructTTS; PromptTTS 2; ControlSpeech	Text descriptions replace specialist acoustic knobs and scale voice/style prompting.	LLMs help annotate or encode prompts, but the model still executes a supplied condition.	The system does not decide what style is appropriate for the interaction.
Context-aware synthesis	2024-2025	PerceptiveAgent; Style-Talker; JELLY; RADKA-CSS; Chain-Talker	User audio, dialogue history, emotion and stored interactions become control observations.	Inference and retrieval begin to choose how the response should sound.	Planning, feedback and environment action remain partial or external.
Agentic speech production	2025-2026	PodAgent; DialogueAgents; ActorMind; AuDirector; AgentSteerTTS	Specialized agents plan content, cast voices, retrieve anchors, synthesize and critique outputs.	Explicit planner/retriever/critic loops make speech generation auditable and revisable.	Most systems produce offline artifacts rather than act in a live environment.
Interactive voice agents	2026	F-Actor; LTS-VoiceAgent; PersonaPlex; ModeratorLM	Control expands from how to speak to when to speak, interrupt, backchannel and assume a role.	Streaming state, incremental reasoning and role-conditioned turn policy enter the loop.	Tool use and long-horizon task reliability are still uneven.
Integrated spoken agents	2026.05-08	DuplexSLA; PRISM; VoxMind; JoyAI-Talker; Harness TTS	Speech, language, tool/action and expressive control share one online decision process.	Reason-plan-act, tool routing and empathic expression close the model-to-agent loop.	Persistent memory, safety, evaluation realism and controllable failure recovery remain open.

Trajectory Synthesis

- **Representation：** latent style tokens → codec/diffusion speech → speech-native and synchronized speech-language-action streams.
- **Control ownership：** explicit user condition → context inference → planner/retriever/critic → online reason-plan-act policy.
- **Temporal coupling：** utterance-level generation → conversational context → streaming/full-duplex speech action.
- **Evaluation：** quality and controllability → context alignment → tool/task correctness → reliability under real-world interaction.

11. Academic Roadmap and Capability Curve

The curve summarizes increasing autonomy, closed-loop control and temporal coupling. Branches expand representative works already listed in this collection; vertical position is conceptual and does not indicate benchmark performance.

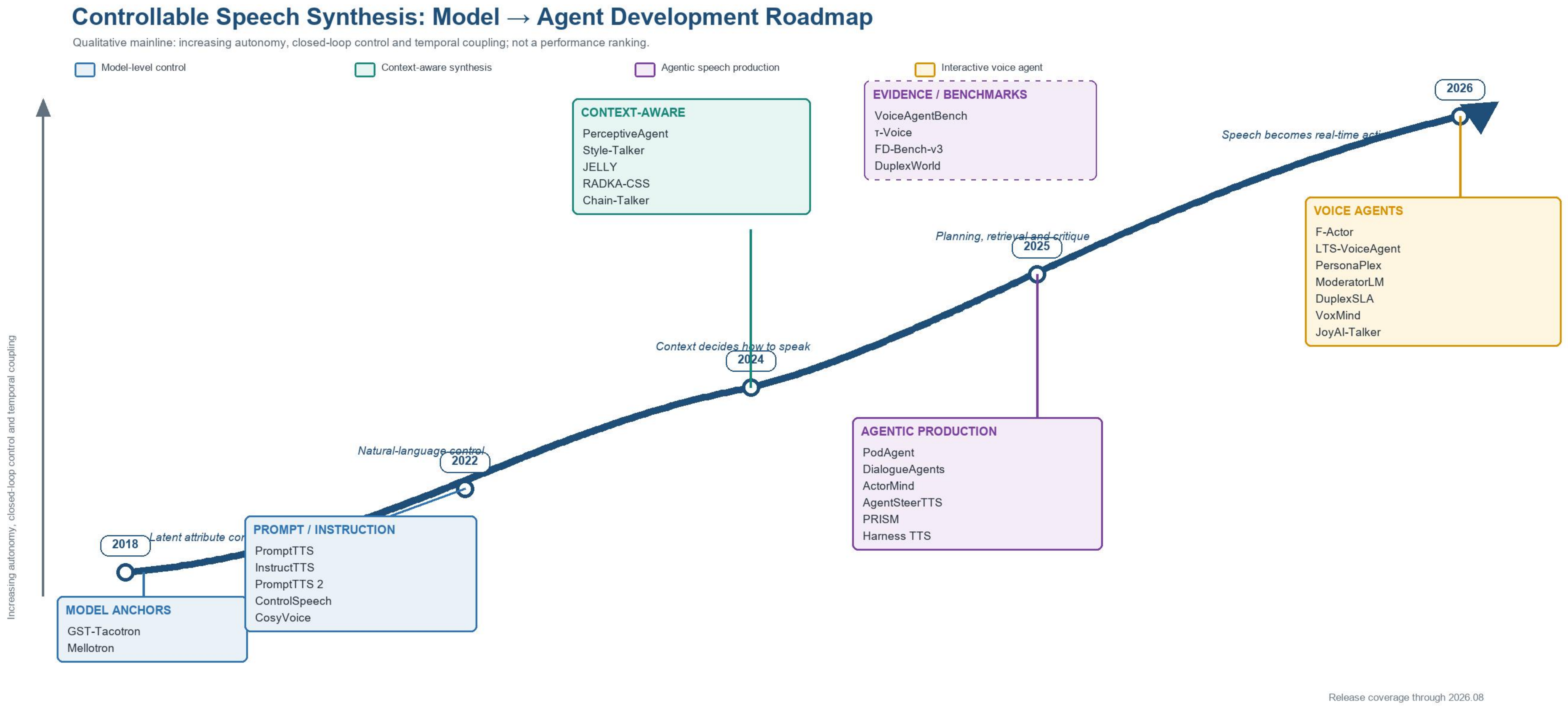

Figure 1. Development roadmap for controllable speech synthesis from model-level control to interactive voice agents. Dashed evidence boxes denote benchmark support rather than model milestones.

Interpretation. 2018–2024 establishes controllable acoustic execution; 2024–2025 moves control inference into the conversational context; 2025–2026 introduces planners, retrieval, multi-agent production and critics; 2026 begins to synchronize expressive speech with online turn policy, reasoning, tools and actions. The main unresolved problem is reliable long-horizon agency: persistent memory, safe expressive control, grounded tool execution and recovery from speech-interaction failures are still weakly unified.